GENETICALLY MODIFIED ORGANISMS AND WORLD OPINION

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A SEMINAR WORK SUBMITTED TO THE DEPARTMENT OF BIOLOGICAL SCIENCES (MICROBIOLOGY OPTION), FACULTY OF SCIENCE, BENSON IDAHOSA UNIVERSITY, BENIN CITY, EDO STATE.
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF BACHELOR OF SCIENCE (BSc.) HONS DEGREE IN MICROBIOLOGY

NOVEMBER, 2018.

CERTIFICATION

This is to certify that this seminar report was presented and compiled by **AGU JENNIFER CHIOMA** during the academic year 2018/2019 under my supervision.

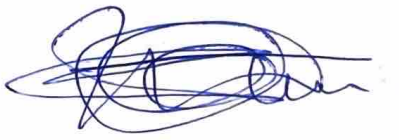


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DEDICATION

This body of work is dedicated to Almighty God, the giver of life for His unconditional love, mercy, guidance and to my parents, Mr and Mrs Agu for their love and support.

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Finally, I would like to thank my friends through peer-review system, made constructive criticisms that enriched my research and eventually my body of work.

PLAGIARISM ATTESTATION

I certify that this report is my own work, based on personal research and that I have acknowledged all material and resources used in its preparation whether they be books, articles, reports, lecture notes and any other kind of document, electronic or personal communication. I also certify that the report has not previously been submitted for assessment elsewhere, and I have not copied in part or whole or otherwise plagiarized the work of other students and/or persons.

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ABSTRACT

Genetic modification involves the insertion of genes that code for desirable traits or removal of genes that code for undesirable traits. Genetically Modified Organisms (GMOs) are organisms (plants, animals and microbes) that have had their DNA altered through genetic engineering.

The controversy that is linked with genetic modification is alarming due to the fact that there are no boundaries or limitations to this form of biotechnology. It is important to determine the pros and cons of this technology while ascertaining the different public opinions towards it. The factors that influence these opinions, the effects of consumer acceptance, international relationships, politics and religion on the implementation of the technology. As at present there are more commercially available GM crops than GM animals, although important developments have been made and several commercial products, such as GM salmons, are awaiting approval.

INTRODUCTION

Genetics has a role to play in every part of life (Daniels, 2018). Genetic modification is a widely talked about topic in the science world as it is both necessary for proper socio-economic development and very controversial. Genetic modification of organisms came to the knowledge of most of the general public in the 1990s and a large number of persons are against the genetic modification of organisms.

Genetically modified organisms (plants, animals or microorganisms) also known as GMOs are organisms that have their genetic material altered by the integration of genes that code for desirable traits like herbicide resistance and pest tolerance. It is understandable why so many persons have a lot of concerns about genetic modification of organisms especially when it involves food crops and livestock. The technology is often called "modern biotechnology" or "gene technology" and foods produced with this technology are referred to as Genetically Modified (GM) foods.

Issues such as the concern of the public for possible hazards due to the consumption of GM food are continuously discussed, but there is always something to add as new discoveries are always made. This report will deal with the controversy concerning applications of genetic modification, public opinion and perception and the politics involved in the implementation of genetic modification. It also reviews opinions which are currently debated in the food biotechnology sector regarding human health and ecology.

HISTORY OF GENETIC MODIFICATION

Humans have been altering organisms for a very long time to give them desirable traits. These desirable traits include better storage capacity, higher yield, durability and resistance to pests. These desirable traits are usually directly or indirectly linked to increasing economic profit of the farmer, company or nation. Our ancestors influenced the genetic material of other organisms by a process called "selective breeding" or "artificial selection", both terms coined by Charles Darwin. Selective breeding was done by breeding two organisms with desirable traits together to keep the traits within the subsequent filial generations.

A summarized timeline of genetic modification indicates the presence of genetic modification, in form of selective breeding during prehistoric times and the Gregor Mendel experiment that occurred in the 1800s to modern biotechnology which gave rise to present-day genetically modified organisms.

TIMELINE OF GENETIC MODIFICATION

The advent of genetic medication can be essentially grouped into four time periods: prehistoric times, 20th century, the 21st century and present day (Smith, 2007).

Prehistoric Times: Since the advent of agriculture 12,000 years ago, farmers have strived to improve their crop resistance to and satisfaction to humans as much as possible.

Over the years, as humans chose certain qualities over others in plants, they moulded crops into what they wanted them to be and what traits they wanted them to express.

20th Century: In the 1800s, Gregor Mendel carried out plant hybridization experiments which gave him recognition as the father of modern genetics. Hybridization involves breeding between organisms of different species although plants have higher ability to hybridize because pollen often disperses onto the flowers of other species. He worked primarily with pea plants between 1856 and 1863, and his work was later drawn upon in genetic engineering.

21st Century: In 1954, Watson and Crick proposed a double helix as the shape of deoxyribonucleic acid (DNA) paving the way for genetic engineering.

1970-a major agriculture company called Monsanto that now controls most of the seed industry, employed chemists to redevelop glyphosate as a herbicide. The Monsanto glyphosate later came to be known as Roundup. Monsanto then went on to become the biggest supplier of glyphosate-resistant crops, known as "Roundup Ready" seeds.

Between 1972 and 1973, U.S. biochemists Herbert Boyer and Stanley Cohen developed a technique that allowed them to cut pieces of DNA in certain places, and attach the pieces to the DNA of other organisms, shepherding in modern biotechnology.

1976-Biotechnology became commercialized, allowing companies to experiment with inserting genes from one species into another for various reasons which could be medicinal, food, or chemical reasons.

1982- First GE plant - A tobacco plant engineered with a yeast gene was created by Mary Dell Chilton, a scientist at Washington University in St. Louis, and Andrew Binns from the University of Pennsylvania (Institute for Agriculture and Trade Policy, 2014).

1988- Scientists inserted genes into soybeans creating glyphosate-tolerant soybeans. Making a crop that was resistant to herbicide made it much easier for farmers to control weeds while producing a high yield. Other GMO seeds such as cotton, rice and tomatoes were developed.

Present Day: There has not been enough research to determine whether GMOs are entirely healthy for humans. One of the concerns includes the fear that altering the natural state of an organism has unforeseen consequences for humans, livestock and the ecosystem as a whole. For now, these concerns have not been substantiated due to the fact that proper researches that will answer these questions have not been carried out.

While the debate over whether GMOs are safe for humans and the environment rages on, perhaps it's important to remember that GMOs aren't necessarily the whole side of the story. Agro-biotechnological companies such as Monsanto and similar companies that sell herbicide-resistant seeds are not exactly leading modern agriculture into a more sustainable era and forcing GMO labels onto food products (or banning GMOs entirely) isn't going to stop them (Mercola, *n.d*).

PRINCIPLES OF GENETIC MODIFICATION

Manipulation of DNA in plants and animals to give favourable products has been carried out for the longest time but with the advancement of technology, direct modification of genetic material is carried out Genetic modification involves alteration of genes; a gene is the basic physical and functional unit of heredity. Genes are made up of DNA and the sequence of DNA carries information that will be expressed as proteins. The alterations of genes give them desirable traits or remove the possibility of expressing undesirable traits coded in the gene. There are many specific ways of genetic modification but the basic idea of genetic modification is that desirable genes are isolated and inserted within the target organism.

1. **Isolation**: the gene that codes for a desirable trait that they want to insert and then change it so that it works in the organism.
2. **Insertion**: They prepare plant cells to be inserted. Insertion is often done using a gene gun, where genes are coated with tiny particles of gold or tungsten with and then shot into a plate of cells.

Alternatively, bacterial infection method can be used to genetically modify plants and it is carried out by infecting plants with the foreign gene from bacteria such as *Agrobacterium tumefecians*. Once the gene gets into the DNA of the plant cell, the cell is cloned (using tissue culture) into a full plant. These steps contain scientific uncertainties and risks for health and the environment (D'Agostino *et al*, 2016).

TECHNIQUES OF GENETIC MODIFICATION

Genetic modification techniques have been used to transfer single gene traits such as herbicide tolerance from soil microbes into plant cells. It has been discovered that genes instead of being constant and static, are dynamic and operate in an interactive system and intertwined with one another. Furthermore, proteins do not function separately; rather they behave in interactive network systems. Gene traits work in the cell by inter-communication and reciprocity. Hence, one gene might not determine one trait, be it herbicide tolerance, or resistance to pest. Therefore, the genetic engineering techniques seem to be imprecise and must include gene optimization steps (Brooker, 2012).

To make a GM crop, the gene of interest is inserted into the crop's genome using a vector. This vector might contain several other elements, including viral promoters, transcription terminators, antibiotic resistance and marker genes. The genes incorporated into a genome, could reside anywhere, and cause mutation in the host genome, and move or rear-range after insertion or in the next generations. Transgenic DNA might break up and reintegrate into the genome again (recombination) leading to chromosomal rearrangement in successive generations and could potentially change the transgenic crops in a way to produce proteins that are allergic or cause other health problems (Hill, 2014). As DNA does not always fully defragment in the digestive system, human gut microflora and pathogens can take up GM materials including antibiotic resistance genes (Braun, 2001). This may cause a reduction of the effectiveness of antibiotics and therefore increasing the risk of antibiotic-resistant diseases. Some scientists have advice have proposed that such markers should be replaced by non-antibiotic marker system in Genetically Modified Organisms production (Chen *et al*, 2010). However, proponents of commercialization of Genetically Modified Organisms

believe that there has not been any evidence of the gene transfer from the food source to gut bacteria.

APPLICATION OF GENETICALLY MODIFIED ORGANISMS (GMOS)

Genetic modification has been researched so it can be successfully implemented in various aspects of society such as agriculture, economy and world hunger.

AGRICULTURE

Today, those who directly see the most benefits from GMOs are farmers and agricultural companies. When farmers plant their crops they generally worry about three things that could prevent a good yield: insects, weeds and weather and implementation of genetic modification provide them with crops that are resistant to these adverse effects or conditions. Such acquired resistance or traits include:

1. **Herbicide Tolerance** - crop can withstand herbicide applications.
2. **Insect Tolerance** - plant produces toxins to kill pest.
3. **Improved Nutrition** – plant produces a substance of nutritive value or is changed to not produce an anti-nutrient.
4. **Disease Resistant** – the crop is resistant to certain diseases.
5. **Stress Tolerance** – the crop is tolerant of stress, low nutrient levels or excess nutrients.
6. **Storage Capacity** – crops can be stored longer to avoid spoilage losses.
7. **Medicinal Uses** – Through this biotechnology, crops are modified to produce medicines or vaccines as products of metabolites.

ECONOMY

The implementation of genetic modification can affect the economy or economic growth in the following ways:

1. **More Food for Less Cost:** The use of genetic modification to improve crop resistance will make crops less susceptible to conditions that cause yield reduction and as such there would be an increased and steady supply of food for the same prices. In time, there may also be economic benefits to farmers who use genetically engineered crops although this will depend on the prices charged by the biotechnology firms for these transgenic crops.

2. **Higher Capital for Food Production:** application of genetic modification affects food production because more capital is required as conventional methods of crop production is replaced with genetic engineering.
3. **Job Creation:** The implementation of Genetic Modification creates work opportunities in the genetics branch of the sciences thereby boosting the development of several science sectors in the economy.
4. **Higher Corporate Control of Food:** Most adversaries of Genetically Modified foods are against it because they feel as though richer countries would control the supply and level of production in developing countries. Most developed counties have been involved in confined field tests of genetically modified foods but developing countries are not well equipped for testing and implementation of genetic modification.

WORLD HUNGER

One can understand the need to feed the world and the people of third world countries but is this benefit so great that it outweighs the risks on the environment and human health (Toft, 2012)

1. **Increased Natality/Population Rate:** As there is a steady supply of food, there would be a rapid increase in population density due to migration and high natality rate. These occur because people prefer favourable conditions provided by ensured food security. There is a need to provide food especially for developing countries and when GM is implemented there will be population increase which can lead to ecological imbalance and increased food requirements.
2. **Death of Fauna:** BT corn and cotton are modified with a gene from the bacterium *Bacillus thuringiensis* and the gene encodes for pest and herbicide resistance (Phillips, 2008). When insects try to pollinate or infest genetically engineered crops they could be harmed by the in-built pesticides.

According to Environmental Protection Agency, GM plants are registered as pesticides although it has been said that that *Bacillus thuringiensis* is used as natural insect control even though it comes to notice that the BT in GM crops cannot be washed off like the herbicide spray. BT cotton caused allergic reactions to labourers working in cotton factories in India (Ashish et al, 2005).

ENVIRONMENTAL MANAGEMENT

The quality of life on earth is linked inextricably to the overall quality of the environment (Kensa, 2011).

Environmental management is a very important aspect of ecosystem sustainability and the green movement as it helps mitigate the effects of man's activities on the environment Genetic modification and genetically modified organisms can be applied in environmental management in several ways, some of which include bioremediation and synthesis of biodegradable plastics.

BIOREMEDIATION

In the process of industrialization, which involves the utilization of natural resources, toxic wastes are released into the environment. This waste is inherently harmful to man and the environment as a whole.

Bioremediation is a technology for removing pollutants from the environment thus restoring the original natural surroundings and preventing further pollution (Sasikumar and Papinazath, 2003). Recombinant bacteria can be obtained by genetic engineering techniques or by natural genetic exchange (transduction) between bacteria, however, ecological and environmental concerns and regulatory constraints are major obstacles for testing genetically engineered microorganisms (GEM) in the field. Due to regulatory risk assessment concerns in bioremediation have largely been confined to the laboratory environment and the uncertainty of their practical impact under field conditions. There are at least four principal approaches to GEM development for bioremediation application. These include:

(1) Modification of enzyme specificity and affinity

(2) Pathway construction and regulation,

(3) Bioprocess development, monitoring, and control

(4) Bioaffinity bioreporter sensor applications for chemical sensing, toxicity reduction and end point analysis.

Genetically engineered microorganisms have a higher degradative ability and have been demonstrated successfully for the degradation of various pollutants under defined conditions. The tested contaminants include chlorinated compounds, aromatic hydrocarbons, heavy metals, and nonpolar toxicants, etc. Development of effective and cost-efficient bioremediation processes is the goal of environmental biotechnology. The combination of microbiological and ecological knowledge, biochemical mechanisms, and field engineering designs are essential elements for successful in situ bioremediation using GEM. (Menn *et al*, 2008).

SYNTHESIS OF BIODEGRADABLE PLASTICS

Some bacteria can produce biodegradable plastics, and the transfer of that ability to microorganisms that can be easily grown in the laboratory may pioneer a movement to make the plastics industry more eco-friendly. In the early 1990s, Zeneca, a British company, developed a microbiologically produced biodegradable plastic called Biopol (polyhydroxyalkanoate, or PHA). The plastic was made with the use of a GM bacterium, *Ralstonia eutropha*, to convert glucose and a variety of organic acids into a flexible polymer (Pusztai *et al.* 2003)

ENERGY CONSERVATION

With genetic modification, food can be grown in areas that are deemed unfit for crop cultivation due to adverse conditions such as drought or excessive rainfall with aids for proper land management and subsequently environmental management.

LIVESTOCK

In the United States, livestock have been consuming feed made from genetically modified crops for almost twenty years. More than two-thirds of GM corn and half of GM soybeans are used for livestock feed. In that time, GMOs have never been detected in the milk; meat or eggs derived from animals fed genetically modified feed. Many studies have been conducted on the potential for GMO DNA or proteins to be transferred into animal tissues. No intact or immunologically reactive protein or DNA has been detected in animal tissue.

Body weight might be significantly altered as it has been shown with the consumption of Mon863 corn and GM rice on rats (Seralini *et al*, 2007).

PHARMACEUTICAL INDUSTRY

This form of biotechnology has enabled researchers to test novel therapies for diseases and to explore the roles of disease outcome. GMOs revolutionized the production of complex pharmaceuticals by enabling the generation of safer and cheaper vaccines and therapeutics. Pharmaceutical products span from recombinant hepatitis B vaccine produced by GM baker's yeast to injectable insulin (for diabetics) produced in GM *Escherichia coli* bacteria and to factor VIII (for haemophiliacs) and tissue plasminogen activator (tPA, for heart attack or stroke patients). Furthermore, GM plants that produce "edible vaccines" are under development.

An edible vaccine is an antigenic protein that is produced in the consumable parts of a plant (e.g., fruit) and absorbed into the bloodstream when the parts are eaten. Once absorbed into the body, the protein stimulates the immune system to produce antibodies against the pathogen from which the antigen was derived. Such vaccines could offer a safe, inexpensive, and painless way to provide vaccines, particularly in less-developed regions of the world, where the limited availability of refrigeration and sterile needles has been problematic for some traditional vaccines traditional vaccination approaches, including HIV/AIDS, tuberculosis, and cancer.

One of the most known genetically modified drugs is GM insulin. Insulin is a peptide hormone that is secreted by the pancreas. It is involved in the regulation of carbohydrate and fat metabolism in the body, it also promotes protein synthesis and glucose utilization. A significant number of diabetics have experienced injurious reactions to GMO insulin, and that some of these have resulted in death. Although GMO insulin poses a risk to some consumers of insulin, its greater availability and lower cost of production make GM insulin more appealing to insulin manufacturing companies.

COMMONLY MODIFIED ORGANISMS

CROPS

There is no nutritional di□erence between GM foods and their non-GMO counterpart unless the nutritional content of the GM crops has been intentionally modified, like high oleic soybeans, or bio-fortified crops, such as Golden Rice.

1. Flavr Savr Tomatoes: modified by the removal of genes responsible for softening of tomato.
2. Bt Corn & Cotton: Bt corn and cotton are modified with a gene from soil bacterium *Bacillus thuringiensis,* the gene encodes for pest and herbicide resistance (Phillips, 2008*)*.
3. The NewLeaf™ potato: a GM food developed using *Bacillus thuringiensis* (Bt), was made to provide in-plant protection from the yield-robbing Colorado potato beetle (Johnson, 2008).
4. Golden Rice: This rice has been modified to have an increased vitamin A level. A due to the incorporation of genes from a microbe and from daffodils(Deepa,2015)

CROP NAME	SCIENTIFIC NAME	TRADE NAME
Alfalfa	*Medicago sativa*	Roundup Ready™
Canola	*Brassica napus*	Laurical™, Optimum® Gly, and Roundup Ready™
Potato	*Solanum tuberosum L.*	NewLeaf™, and Innate™
Tomato	*Lycopersicon esculentum*	FLAVR SAVR™

Table 1: some commonly modified and registered GM crops

MICROBES

Genetically modified microbes are referred to as Genetically Engineered Microorganisms

1. *Pseudomonas putida* KT2442: the metabolic pathway has been modified for hydrocarbon degradation (Panke *et al.*, 1998).
2. *Sinorhizobium meliloti* : This species of microorganisms are modified by the addition of nitrogen-fixing genes to boost their nitrogen fixation.

SOCIO-ENVIRONMENTAL AND POLITICAL EFFECTS OF GENETICALLY MODIFIED ORGANISMS

There is a connection between one's social background and their political values because their political ideals are influenced by their environment and their social interactions. Though a number scientists have said that GM foods are safe, the debate over its safety continues and will continue for a while because the controversies of Genetic modification and its implementation extend into social and political controversies as this form of biotechnology affects every facet of society and the environment

SOCIO-ENVIRONMENTAL EFFECTS OF GENETICALLY MODIFIED ORGANISMS

Potential benefits notwithstanding, hypothetical risks associated with genetic modification and genetically modified organisms have fuelled controversy, especially in the food industry. Implementation of genetic modification can result in any of the following negative environmental consequences:

1. **Creation of Superbugs**: As pesticides are applied to insects that infest crops, some of the insects withstand and survive the application. Each subsequent generation becomes more resilient to the pesticide. This leads to the additional application of pesticide or the use of an even stronger pesticide (Paoletti, 2001).
2. **Antibiotic resistance**: One of the ways that crops are modified to be more resilient to disease is to artificially place antibiotic genes within the DNA of the crops. If this property were transferred to a person eating the food, antibiotics might not have the usual effects against infection. This could be a contributing factor to the "super bacteria," such as Methicillin-resistant *Staphylococcus aureus* (Perkins, 2017).
3. **Hazardous to soil**: Crop residues may pose a potential hazard to above ground and soil organisms (flora and fauna) a□ected by containing genes that code for herbicide production and soil leachate contains herbicide which could reduce soil integrity by depleting nutrients in soil and population of microbes in the soil (Soil Association,
4. **Use of GM foods as feed for livestock**: In addition, if GM foods are not considered safe for human consumption why should they be approved for animals? Humans can inadvertently consume foods that contain GM products fed to animals, i.e., crops modified for enhanced productivity in animals. One should not forget that every single GM food through the food chain will eventually reach human beings. (Arvanitoyannis and Dona,2009)

5. **Helping Reduce Greenhouse Gas Emissions:** As herbicide-resistant corn and soybeans were introduced in the U.S. Genetically modified crops with herbicide-resistant traits can allow farmers to manage weeds wit on the traditional method of tilling fields. Generally, fewer tillage results in fewer tractor-related emissions. (Goode, 2015)
6. **Exposure of consumer to the viral infection**: there is a concern that the existence of viral promoters in the vectors carrying the foreign genes might expose the consumer to the viral infection. (Myhre *et al*, 2006)
7. **Spreading Prosperity to Growing Nations:** Since the introduction of herbicide-resistant maize, farmers in South Africa have experienced notable gains in inefficiency. GM crops have helped improve yields and allowed growers (on smallholder farms in particular) to adopt intensive practices—like hand weeding. If this trend continues, GM crops will help create more economic security developing nations.
8. **Formation of Non-Governmental Organisations**: Due to the high rate of opposition on the topic of genetic modification and genetically modified foods, NGOs may be formed to counter the decisions of the government. This could lead to a segregated society.
9. **Legal implication for farmers and scientists**: Companies that produce GM seeds exert broad control over the use of seeds by mandating that the farmers enter into a "Technology Stewardship Agreement" which outlines the conditions under which the seed may be used, where it may be grown, where it may be sold, and the brand of herbicide that may be used and it also restricts the research on seeds. Scientists are no longer free to conduct independent tests on the seeds. They need to obtain approval from the seed company or gene patent holder (Deepa, 2015).

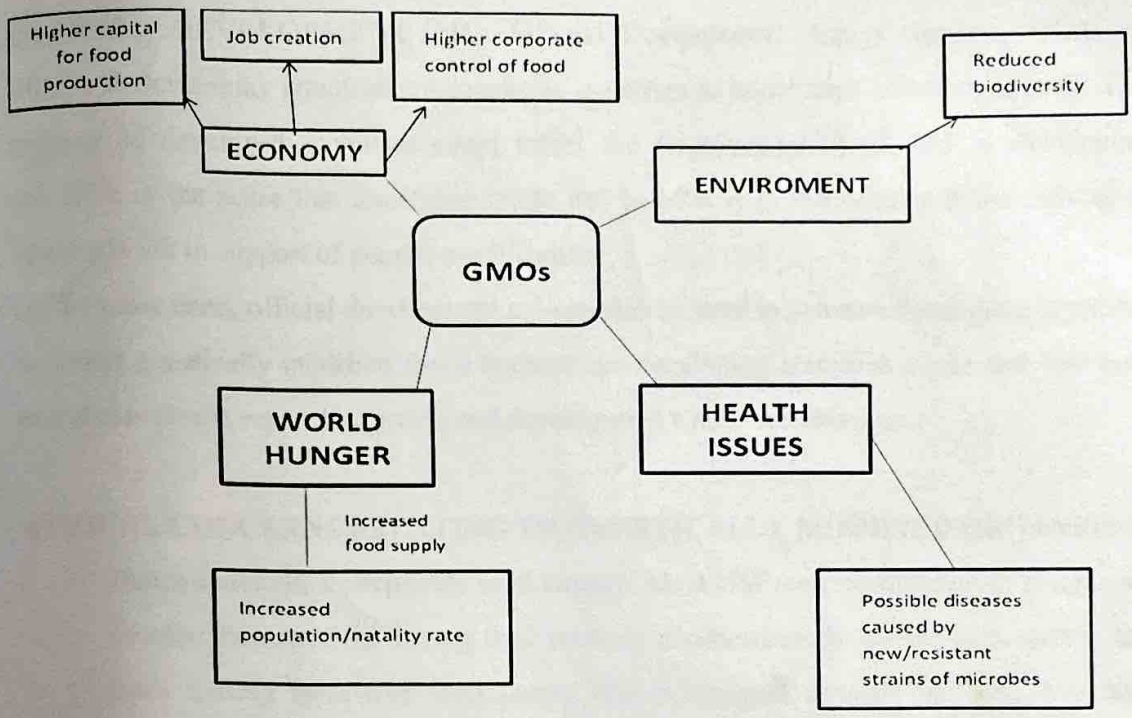


Fig 1: Socio-Environmental Effects of Genetically Modified Organisms

Source: (Anonymous, 2018)

POLITICAL EFFECTS OF GENETICALLY MODIFIED ORGANISMS

There is a relationship between agriculture, government policy, and the environment, therefore, food choice is also influenced by policies structured by international agreements and culture surrounding the people. The issues of labelling and Official Development Aid have impacted the implementation of Genetic Modification as they both have both incited political outcomes.

LABELLING: The controversy of labelling has affected trade relationships between several countries especially between the ones that have strict laws about labelling like the European Union (EU) and those that do not require strict labelling procedures like the United States. In the early 2000s, all foods and animal feed in the EU were required to be labelled if they consisted of or contained GM products in a proportion greater than 0.9 percent (Britannica, 2014). The strict regulations on GM products in the EU have been a source of tension in agricultural trade. In the late 1990s, the declaration of a moratorium on the import and use of GM crops led to trade disputes with other countries, particularly the United States and as a result, the European Union implemented regulatory changes that allowed for the import of certain GM crops.

OFFICIAL DEVELOPMENT AID: Official Development Aid is financial assistance offered to developing countries by developed countries to boost their economic growth. The opinion of developed countries could affect the implementation of GM in developing countries in the sense that assistance might not be offered to the country if the developed country is not in support of genetic modification.

On the other hand, official development aid can also be used to pressure developing countries to accept genetically modified foods because the developing countries might feel like they would experience economic growth and development without assistance.

ETHICAL CONCERNS RELATING TO GENETICALLY MODIFIED ORGANISMS

1. **Dependence on a corporate seed supply**: Most GM seed manufacturing companies prohibit farmers from saving their on-farm produced seeds for the next season and from sharing them with their peers. This is imposed through elaborate contracts, agreements, and conditions, which are imposed by the multinational GM seed companies.
2. **Destruction of cultural norms**: According to Makanya (2004) seed sharing (with neighbours, relatives and friends) is a cultural norm in many African communities and the introduction of GM seeds will jeopardise these traditional and vital practices (Makanya, 2004).
3. **Lack of proper toxicity tests**: The possibility that an unidentified compound may be present in the GM food makes crucial that each transgenic food as whole food and not as a single protein should be tested directly for toxicity in animals
4. **Cross-contamination**: The modification of the genome of an organism gives inherent advantages over non-modified organisms due to natural selection. There is great uncertainty regarding the products or possible outcome of cross-pollination between modified organisms and non-modified organisms.
5. **Possible reduction of biodiversity**: This is as a result of modified organisms having an added advantage and if they are allowed to reproduce, there is a risk of wild-type organisms being dormant and subsequently overrun but if cross-breeding can actually occur there will be filial generations with desirable traits gotten from the modified organisms. This is regarded as possible genetic pollution.
6. **Tampering with nature**: Many people especially religious people and affiliated religious bodies often have a strong stance against genetic modification as they feel as

though it interferes with the natural order of the things and the proponents are "playing God"

HEALTH CONCERNS RELATING TO GENETICALLY MODIFIED ORGANISMS

Our food is an important determinant of our health because whatever we ingest is metabolized in our system and creates physiological resections which maybe extreme regarding the ingested substance. The possibility that an unidentified compound may be present in the Genetically Modified food makes crucial that each transgenic food as whole food and not as a single protein should be tested directly for toxicity in animals; although as (Kuiper *et al.* 2004) state there are limitations in establishing dose-response relationships.

It has been said that there are no harmless substances only harmless ways of using them (Enabulele, 2018). There is a high possibility of GMOs being carcinogenic due to the mutagenesis that occurs during modification. There are also naturally occurring substances like potatoes that contain that has lower levels of asparagine, which when heated, turns into acrylamide which is a potential carcinogen.

The Potential for Pleiotropic and Insertional Effects

Pleiotropy is the influence of a single gene on multiple phenotypic traits. This interaction of the newly added gene (transgene) with the activity of the existing genes and biochemical pathways of plants, may lead to disruption of metabolism in unpredictable ways and to the development of new toxic compounds.

According to Arvanitoyannis and Dona, genes account for only a part of the control of the biochemistry of organisms, and organisms have a level of control above genes that interact with genes explaining why genetic engineering is so unpredictable, with different results produced by each attempt and why the products are often unstable.

Effects on the Gastrointestinal Tract: Pusztai *et al* (2003) stated since it is the genetic manipulation process itself which led to toxicity, similar hazards might be seen in animals or humans fed genetically-manipulated soy, canola, and corn over a long period of time

EFFECT OF GENETICALLY MODIFIED ORGANISMS ON THE ECOSYSTEM

The environment is extremely important as it encompasses both living and non-living things. The environment is to be protected and safeguarded at all cost and every possible threat to the environment or changes in conditions that may be damaging to the ecosystem should be assessed and rectified as a means of environmental management.

The implication of genetically modified organisms is that biodiversity could be compromised as GM crops are more likely to withstand harsh environmental conditions and due to natural selection they would be prevalent in the environment thus leading to competition and the subsequent dying out of wild-type species. Reduced biodiversity of fauna, flora and soil nutrients can also be as a result of unprecedented effects of cross-contamination as a result of pollination between GM crops and wild-type crops (Glass, 2013).

Traditionally, farmers practised polyculture, or the planting of different crops on a rotational basis and another point of interest is the increased monoculture which is the planting of only one variety of crops in a farmland. This is because of the corporate bodies involved in seed commercialization release only a particular variety of crops to farmers. Monoculture has led to staggering effects on our agriculture and environment as farmers are forced to use more toxic pesticides and herbicides against pests and weeds, which in turn form resistance against these chemicals (superweeds) which then requires agribusiness industries to produce more toxic pesticides, resulting in a never-ending cycle of pesticide usage. (Mercola, n.d)

PERCEPTION OF GENETICALLY MODIFIED ORGANISMS AROUND THE WORLD

Perception of any product is greatly influenced by opinions of the members of the public which in turn is influenced by their lifestyles, values and morals, religiosity, food habits and stance on other factors that influence manufacture of that particular product.

PUBLIC OPINION

Public opinion is the accumulation of individual opinions on issues of concern to the public. Public opinion is especially important in a democratic society as public opinion determines the state of the nation; this came about as a result of urbanization and socio-political influences (Bawa and Anilakumar, 2012).

The success of a good government is dependent on policies that precede national growth and international relationships and the ability of the government to appeal to the masses as their opinions should mould action of government. Policymakers learn of public opinion in a number of ways including polls although several factors, such as sampling error, systematic errors and lack of participation can make interpretation of public opinion difficult.

Consumer's opinion towards genetic modification and genetically modified organisms vary from nation to nation and even within the same nation from one demographic to another.

IMPORTANCE OF PUBLIC OPINION ON GENETIC MODIFICATION IMPLEMENTATION

Biotechnology is recognized as one of the most innovative technologies to enhance food security, economic growth, human health, socio-political relationships, and agricultural trade. Knowledge about biotechnology in general and genetically modified foods play a role in determining consumer's perceptions of risks and benefits and subsequently attitude towards genetic modification.

General Importance of Public Opinion

1. **Encourages Public Participation**: As individuals express their opinions on policies and laws of their government they are indirectly participating in the governance of state thus reducing political apathy.
2. **Shapes government image**: level of importance of public participation in policy making in a government is an indicator of whether the government is democratic or not. A democratic nation needs to be aware of the demands and supports coming from the public.
3. **Accountability of government**: The people can demand the government to account for their activities while in power and these demands compel the government to be accountable.
4. **Public Sensitization**: One of the functions of the mass media is to educate their audience and readers. As opinions are expressed on various issues at these various media outlets, the masses get educated government policy directions and programme to develop the country.

Importance of Public Opinion on Genetic Modification Implementation

Forming an opinion of Genetically Modified Organisms and Genetic Modification requires considering many controversial topics, such as corporate control, world hunger, human rights, moral boundaries, the validity of scientific research, government and international, and GMOs' role in these issues. Consumer acceptance is conditioned by the risk that they perceive from introducing food into their consumption habits processed through technology that they hardly understand

1. **Misunderstanding of scientific research by the masses:** one of the challenges faced by GM researchers, companies and scientists is that a lot of GM research is miscommunicated to the general public leading them to form mostly negative

opinions on the technology. For example, the implementation of genetically modified crops in third world counties is seen as a lifesaver by some persons while others interpreted it as a form of modern-day colonialism.

2. **Slows down research:** An unwilling audience or population is rather discouraging for researchers and scientists because it would seem like their life's work is futile or inherently harmful to the masses. Some scientists have bowed out from GMO research due to the increasing backlash they received over time. This leads to stagnancy in research of genetic modification and genetically modified organisms (Kuiper *et* al. 2004).
3. **Middle ground opinions**: Some persons and groups are in support of genetic modification for other purposes but not for food purposes because they see it as profiteering. Forming a middle ground opinion is seen as some form of progress because the two extremes, opponents and proponents sometimes find it difficult to pass across their claims and empirical proof to support their argument. Another middle ground opinion is the support of GM foods but that patented seeds and corporate control by big companies like Monsanto should be excluded because the links between biotech industries and food security are complex and multi-faceted (Hallman *et al.*2004)
4. **Policy Creation**: Public opinion of GMOs plays a large role in policy creation, regulation and development of new technologies. Recognizing that a spectrum of perspectives on GMOs exists, exempts decision makers from the pressure to choose a side of the debate or adhere to the desires of powerful interest groups, and this allows critical thinking and productive conversation to exist in the public and political spheres.
5. **Halts funding of research**: Due to overall negative opinions on genetic modification, some corporations may lose funding from donors. This also applies to independent researchers and scientists

PERCEPTION OF GENETIC MODIFICATION IN DIFFERENT PARTS OF THE WORLD

Perception of Genetic Modification and Genetically Modified Organisms differs from continent to continent and it is influenced by age, religion, race, level of education and development in member states of that continent.

NORTH AMERICA

The United States of America and Canada are the major producers and exporters of GM crops and products. In the United States, precaution is rarely stated explicitly in any of its laws regarding GMOs and the lack of strict labelling rules is a source of controversy. In all, 85 GM foods have been approved for sale in Canada (Smith, 2007). Although the opposition to this form of biotechnology is still present, the rate of acceptance of Genetic Engineering and Genetically Modified crops has increased in the United States over the last few years (Fig 2) due to rigorous sensitization

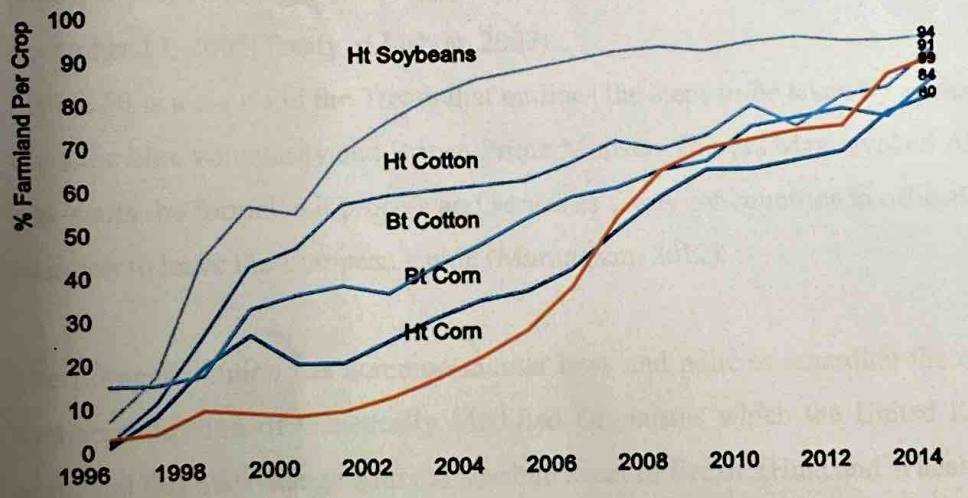


Fig 2: Adoption of GMO crops in USA, 1996-2014

Source: USDA Economic Research Service

EUROPE

Europe is one of the largest importers of GM foods, GMOs are not grown in Europe and European Union countries but their importation is allowed as long as products are duly labelled. (Murnaghan, 2012).

Effects of Brexit on GMO Policy in the United Kingdom

The European Union is a supranational organisation consisting of 28 member states (Austria, Italy, Belgium, Latvia, Bulgaria Lithuania, Croatia, Luxembourg, Cyprus, Malta, Czech, Netherlands, Denmark, Poland, Estonia, Portugal, Romania, Finland, France, Slovakia, Germany, Ireland, Slovenia, Greece, Spain, Hungary, Sweden, United Kingdom) created to create better economic and political connection in Europe.

The exit of the United Kingdom from the European Union is referred to as Brexit was triggered by Article 50 of the Lisbon Treaty. The Lisbon Treaty is an agreement that updated regulations for the European Union, establishing a more centralized leadership and foreign policy, a proper process for countries that wish to leave the Union, and it was signed on December 13, 2007(Treaty of Lisbon, 2007).

Article 50 is a clause in the Treaty that outlines the steps to be taken by a country seeking to leave the bloc voluntarily and British Prime Minister Theresa May invoked Article 50 which kick-starts the formal exit process and serves as a way for countries to officially declare their intention to leave the European Union (Murnaghan, 2012).

The European Union has common market laws and policies regarding the importation and commercialization of Genetically Modified Organisms which the United Kingdom had to adapt and this may change with the possible event of Brexit. (Hunt and Wheeler, 2018)

ASIA

Two of the largest producers of transgenic crops in the developing part of the world are India and China although the first GMOs in Asia were cultivated in the Philippines in 2002 (Cornish, 2018). Insect resistant cotton is the most commercially produced transgenic crop in Asian countries.

Various Asian countries are in the process of establishing their legislative framework for commercial release of GM crops. India has established a committee known as the Genetic Engineering Approval Committee to oversee GM crops application. Regulators must be able

to recognize acceptable GM crops and propose changes to existing GM farming practices to obtain the balance that society demands (Singh *et al*, 2006).

AUSTRALIA

Although importation of GM crops is allowed, only three GM crops (canola, cotton and safflower), are cultivated whereas others are undergoing controlled field tests. (SBS, 2013).

All GM crops grown or imported in Australia (commercial and experimental) are approved by the Regulator only after a scientific assessment shows that they are safe for people and the environment.

AFRICA

Africa is prone to harsh climates and implementation of GM crops helps boost food security and income generation. Cultivation of GMOs is already carried out in South Africa, Burkina Faso, Sudan and Egypt while Uganda, Cameroon and Ghana have conducted/are conducting controlled field tests. Biotechnology can provide farmers with increased crop productivity and fresher produce.

Different African countries have differing opinions and attitudes towards Genetic Modification and Genetically Modified Organisms (Fig.3), this is as a result of varying policies and public perception from region to region.

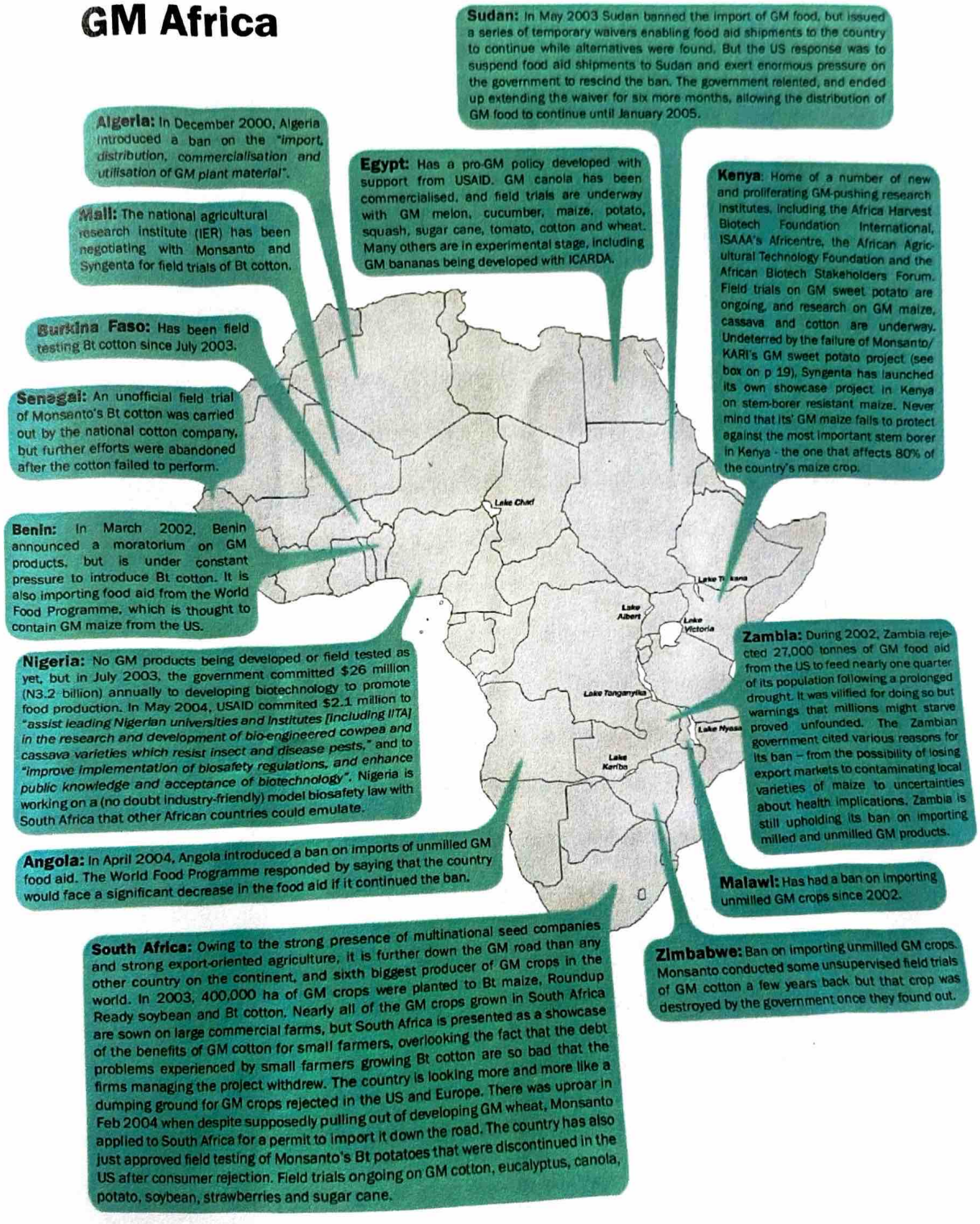


Fig 3: GM map of Africa

Source: Makanya (2004)

GENETICALLY MODIFIED ORGANISMS IN NIGERIA

Nigeria officially signed the Biosafety Bill into law in 2015 by the former president Goodluck Ebele Jonathan, making it eligible to join the group of nations that are using genetic engineering to boost food production. In an interview with Onyeji Ebuka, Rose Gidado who is the Scientific Officer and Assistant Director of the National Biotechnology Development Agency (NABDA), she said Nigeria is fully in support of this form of biotechnology as two agencies, the National Biosafety Management Development Agency and National Biosafety Management Agency, have been created for the research, awareness creation and policy formulation on GMOs and regulation respectively.

Section 33.A of the National Biosafety Management Agency Act , indicates the regulations to be enforced for genetically modified crops and animals and risk assessment tests to be carried out but it would take a while for any appreciable effect to be made from the Bill due to lack of infrastructure and personnel (Odjajare, pers. Comm., 15 Nov).

Despite the signing and implementation of this Act, many Nigerians are firmly opposed the movement mostly due to religious beliefs and this led to a formation of Non-Governmental Organizations such as Health of Mother Earth Foundation (HOMEF).

The Director of this foundation in Nigeria, Ninimo Bassey is vocally anti-GMO and he has stressed that there are serious challenges GMOs pose in the areas of toxicology, allergy and immune dysfunction and genetic disorders which make it very important that Nigeria adopts the precautionary principles such as labelling and the way to improve the economic situation for farmers is to invest in organic agriculture, provide farmers with extension services, needed infrastructure, good roads and access to land and loans. Support for farmers should include investment in research and exploration of agroecology approach to the problems of pests and diseases (Onyeji, 2018).

CONCLUSION

Advancements in science are usually seen as good despite the difficulties and controversies that comes with communicating the technology to the masses. Genetic Modification is one of those topics where diverse knowledge is required to form an opinion regarding it. The on-going debate about GMOs is important as their incorporation in the food and agriculture industry seems beneficial but more long-term studies are needed to analyze the safety and risks which are of great importance. Before condemning GMOs it is important to consider if the negatives of its implementation outweigh the positives. The importance of labelling and public awareness about Genetic modification and genetically modified organisms to ensure that the public has a choice on whether or not to eat such crops mainly due to the country's socio-cultural and economic realities and so as to not infringe on their basic human rights such as the right to choice by enforcing the use of Genetic Modification on them.

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